

LESSON PLANS FORMAT (For Science)

Name: Seanne Clemente	Date: November 29, 2023
Title: Statistics with Skittles (and More!)	Grade/Level: Middle School

Rationale: *Why did I select this lesson? Why is it important to students? How does it build on previous learning?*

Scientific knowledge is almost always based on data collected from experiments and the statistics used to make sense of that data. Understanding data in its various forms & recognizing that its interpretation is variable and prone to misunderstanding is key to understanding the scientific process and its limitations.

Approximate (Amount of) Time Required for Lesson: A single lesson can range in length, from one period to several days. If your lesson extends beyond a day, please indicate it in the lesson plan by dividing it into sections for each day, along with the approximate amount of time (e.g., Day 1 – 30 minutes).

One period: 90 to 120 minutes.

Measurable Content Objective(s): *What do I expect students to know, understand, or able to do as a result of this lesson? Describe the objectives in measurable actions.*

Sample sentence frame for measurable objective (Science): Explain (action/scientific practice) that an objects' density determines whether it floats or sinks (content knowledge).

Identify various forms of data in their everyday lives, such as their height or the number of TikTok videos they watch daily.

Apply various measures of central tendency in real-life situations, such as measuring the average number of orange Skittles in a bag, or the number of flowers a bee visits.

Explain why median and mean are different, and why one or the other would be better at explaining a phenomenon, such as average income.

Related Prior Learning: If this lesson builds on concepts or skills learned prior to this lesson, briefly summarize those here. This is especially important if students are expected to refine their ideas/models based on previously learned principles. This part should also reflect prior lessons in this unit.

Students should be familiar with the scientific process (observations, hypothesis forming, experiments) and the role data plays in it. Students at this level are usually familiar with measures of central tendency, albeit at varying stages of proficiency.

[OPTIONAL] Massachusetts Curriculum Frameworks/Common Core State Standards

(discipline, standard number, and description): For science standards, make sure to specify how the performance expectations are supported by knowing and doing as reflected by the three dimensions of the NRC framework (disciplinary core ideas DICs, science and engineering practices SEPs, and cross-cutting concepts CCCs). [Check NGSS @ NSTA for further support](#)

Attending to equity: *How are you intentionally bridging students' lives and classroom learning, as well as attending to students' sense-making repertoires? How are you designing learning environments that are responsive to a large range of abilities?*

- Using situations and scenarios that are immediately familiar to students – such as watching TikTok videos
- Allowing students to provide their own examples, and share from their own experiences
- Making the formal assessment a group activity and checking in regularly on all groups
- With the income inequality activity, allowing students to recognize the potential consequences of unfair data representation, using a situation that is most likely relevant to them

Methodology: *How will I guide students toward the objectives?*

- Materials
 - Projector and laptop for presentation
 - Worksheets
 - Writing materials
 - Calculators
 - Fake money (e.g. Monopoly money)
 - 2.13 oz. bags of Skittles candy
- Differentiation—*clearly reflecting individual students IEP goals/objectives*
- Strategies for promoting a productive and collaborative learning community.
 - Interspersion of small activities during the presentation to encourage student participation
 - Allowing students to work in pairs or small groups
 - Checking in with students regularly during the Skittles activity

Worksheets, Handouts, and Additional Resources: *Such as books or websites that could enhance/complement instruction and/or activities.*

- A worksheet will be provided for students during the Skittles activity

Assessment of Student Learning: *What evidence will demonstrate to me (and the students) that the lesson objective(s) have been met?*

Formative (informal) Assessment: *How will you determine if your students are learning during the lessons? Include how students will demonstrate their learning, what you will look for to understand what they have learned, and how you will provide feedback.*

- Instructor will occasionally ask students to discuss their ideas and/or provide examples.
- In all stages of this module, a small activity has been designed to reinforce the main idea.

Summative (formal) Assessment: *How will you determine if individual students have acquired the knowledge, skills, or strategies you expected them to learn? Include how students will*

demonstrate their learning, what you will look for to understand what they have learned, and how you will provide feedback.

- The final Skittles activity is an evaluation of whether students can independently calculate measures of central tendency learned in this module. Students will turn in their worksheets after participating in the Skittles activity; following this the instructor will validate calculations.

Instructional Strategies and Learning Tasks:

Pacing	Teacher Actions	Student Learning Activities		Form of Assessment
<u>Introduction (x minutes)</u> Most lesson plans start with some kind of introduction, even if it is very brief, during which the teacher sets the stage, activates prior knowledge, and/or introduces the strategy/skill/content for the lesson.				
5 mins	Introduction, Recap of last week	Prepare for module by reflecting on last week's activities		N/A
<u>Learning Tasks/Activities (y minutes)</u> This might be subdivided into sections for specific tasks. Checks for understanding (related to the objectives) can be included throughout the lesson (typically via questions).				
10 mins	<i>What is data?</i> Presentation "Price is Right" activity	<u>Presentation:</u> identify what information counts as data, and why data is important <u>Activity:</u> explore different kinds of data in everyday situations (ex: the amount of ice cream consumed by an average person, or the number of TikToks watched per second)		Questions during presentation Participation in activity
15 mins	<i>Interpreting data I</i> Demonstration: calculating central tendencies	Learn how to calculate mean, median, and mode using three scenarios (pollinators/flowers). Practicing these independently.		Practicing calculations of mean, median, and mode
15 mins	<i>Interpreting data II</i> Mean vs. median discussion Income inequality activity	<u>Discussion:</u> Talk about why some central tendencies are sometimes more accurate at describing a phenomenon <u>Activity:</u> Use Monopoly money to demonstrate why means can be misleading when some		Participation in activity

		people get much more money than others.		
20 mins	Skittles Activity	Practice skills learned in module by analyzing the flavor distribution in a bag of Skittles.		Participation in activity, validation of worksheet
<u>Closing (2 minutes)</u> A lesson also typically ends with some kind of closure during which the teacher restates, clarifies, and consolidates the objectives, that is, summarizes the learning for the lesson.				
5 mins	Conclusion	Summarize day's main ideas, opportunity to ask final questions		N/A

Helpful Guidelines:

- Include questions you would use when leading discussions, written *verbatim* (exactly as you would ask them), and set off questions with bullet points; make a bullet point list if more than one question.
- To lead a true "discussion," questions should be relatively open-ended, not "one correct response."
- Questions should be followed by anticipated responses. Even when questions are primarily for "brainstorming," you can include examples of responses you can anticipate from students.
- In some cases, it is also clearer if instructions and/or explanations are written *verbatim* or "scripted;" if this is done, also "offset" these in some manner and/or use quotation marks to indicate "scripting."